

TITLE PAGE

- ❖ **Problem Statement ID** – **SIH26084**
- ❖ **Problem Statement Title** – Real-Time Convective-Scale Nowcasting (0–6 h) of Severe Storms using Multi-Source Data Fusion
- ❖ **Theme** – Disaster Management / Smart Automation
- ❖ **PS Category** – Software
- ❖ **Team ID** – **85508 (SIH26084_BUMBLEBEE)**
- ❖ **Team Name** – **Bumblebee**
- ❖ **Ministry / Agency** – Ministry of Earth Sciences (MoES) & India Meteorological Department (IMD)



GIST : BUMBLEBLE fuses radar, INSAT-3DR & lightning data on a 2 km grid to nowcast severe storms 0–6 h ahead, with live arrival countdowns.

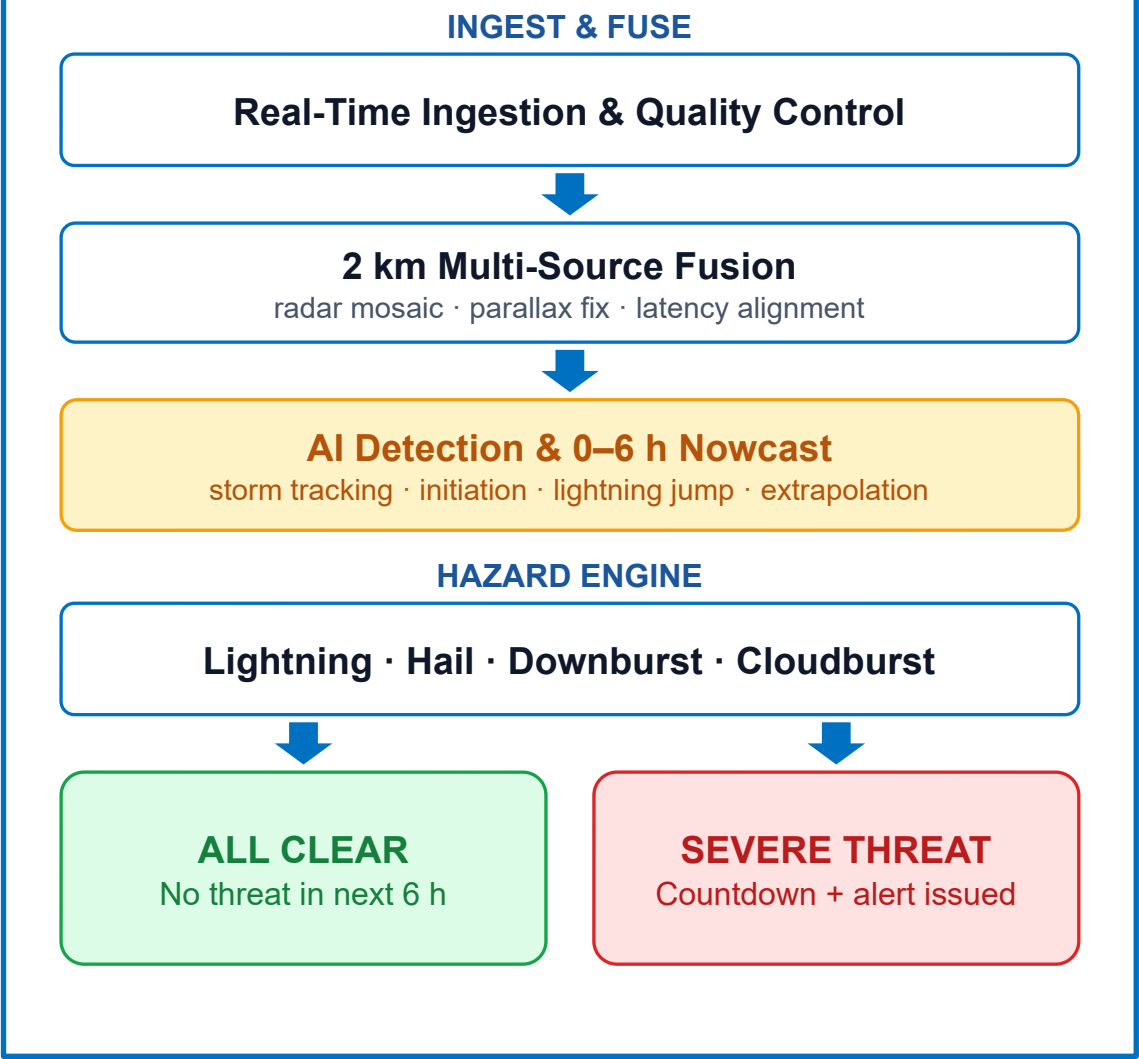
IDEA / SOLUTION

- ❖ **Multi-Source Data Fusion** – DWR radar, INSAT-3DR IR/WV & lightning on one 2 km grid
- ❖ **Early Storm Detection** – cloud-top cooling flags new storms before radar sees them
- ❖ **0–6 h Hazard Nowcast** – lightning, hail, gust & cloudburst every 10 min
- ❖ **Live GIS Dashboard** – arrival countdowns for 80+ cities, airports & farms

PROBLEM RESOLUTION

- ❖ **Lightning: India's #1 natural-disaster killer** → minute-level shelter alerts
- ❖ **Storms form in minutes** – too fast for coarse NWP; we nowcast at 2 km
- ❖ **Radar gaps** are filled by a satellite + lightning proxy
- ❖ **Vague district warnings** replaced by minute-level ETAs per place

BUMBLEBLE NOWCAST FLOW

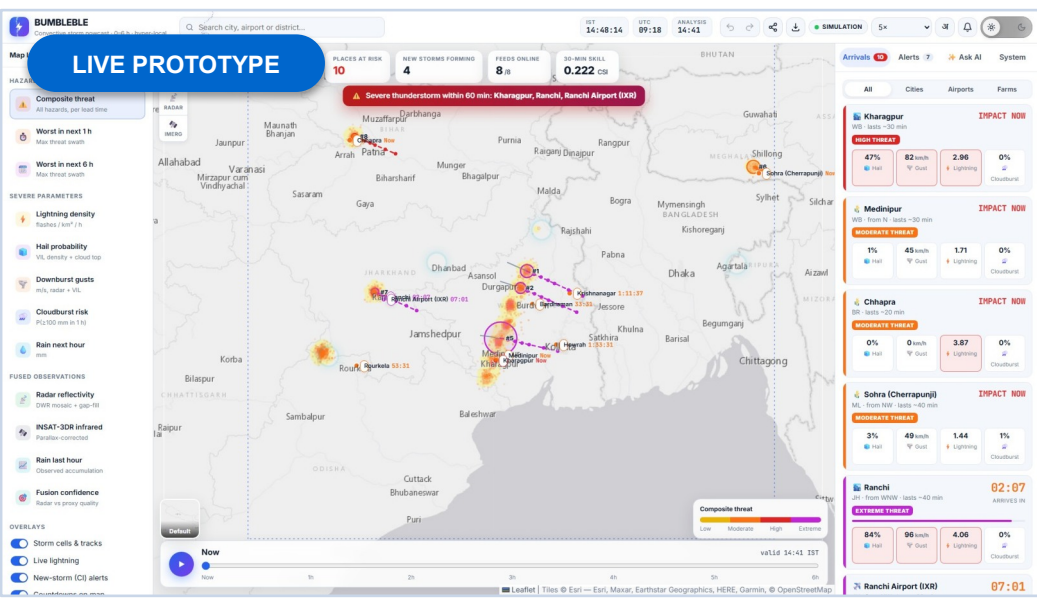


Live Demo

<http://localhost:8000> · one-click Docker cloud deploy

UNIQUENESS

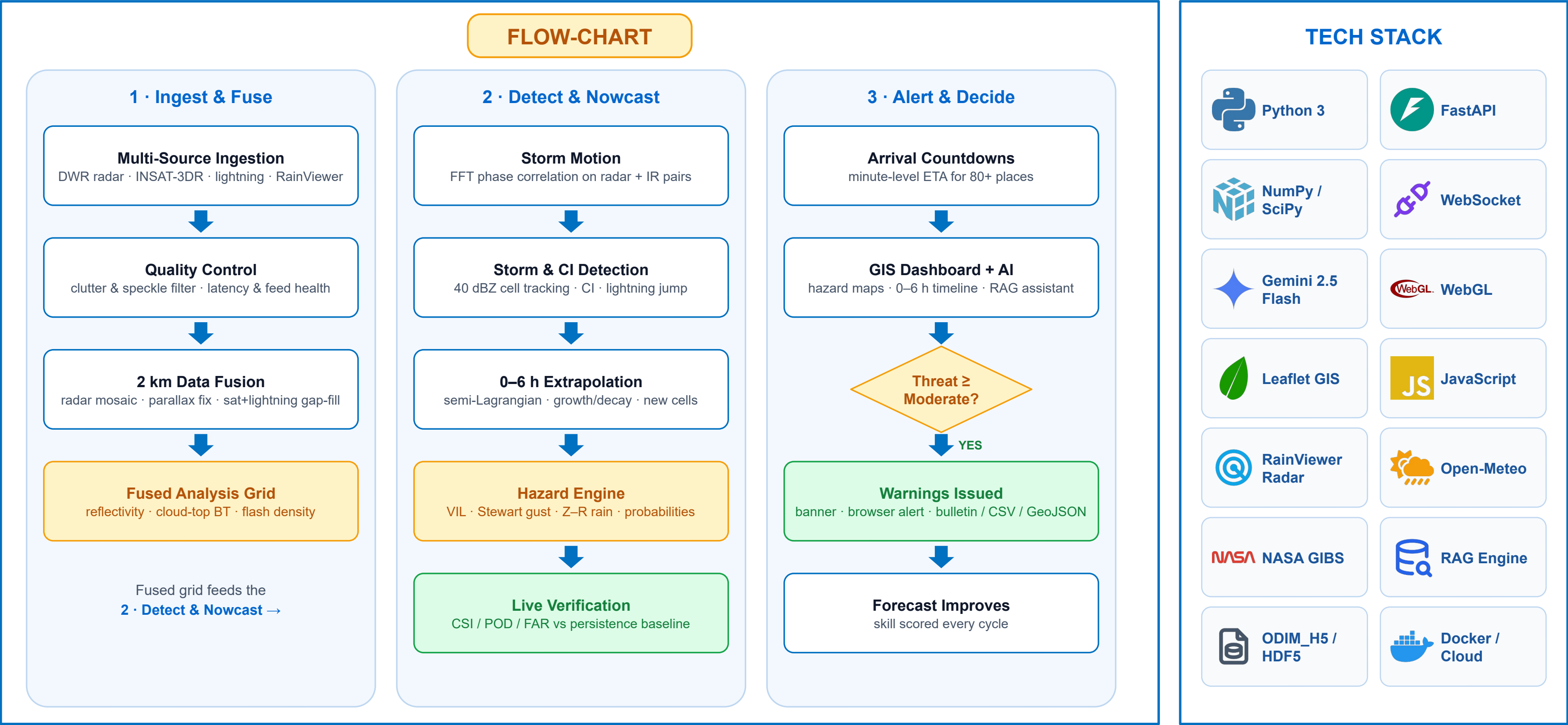
- ❖ **Minute-level countdowns** for 80+ places
- ❖ **Outage-proof:** satellite + lightning fill radar gaps
- ❖ **Self-verifying:** skill scored live vs baseline
- ❖ **AI + RAG assistant** (Gemini) in EN / हिन्दी



Live dashboard · simulated Nor'wester storm

Prototype Status: Working

9/9 automated tests · 150-min storm benchmark
Realtime (live radar) + Simulated Nor'wester demo



ANALYSIS OF FEASIBILITY

Economic

- ❖ **Near-zero data cost:** free RainViewer, Open-Meteo & NASA feeds today
- ❖ **Funding:** MoES / IMD, NDMA & state disaster-tech programmes
- ❖ **Low infrastructure:** one CPU container — no GPU or supercomputer
- ❖ **Avoided losses** → lives, crops, flights & power lines protected

Technical

- ❖ **Working prototype:** 9/9 tests pass; runs on live radar
- ❖ **<15 s per cycle** for all 37 lead times on a laptop CPU
- ❖ **Containerized** (Docker) → free cloud deploy (HF Spaces)
- ❖ **IMD-ready:** ODIM_H5 radar, INSAT HDF5 & lightning push API

Scalability

- ❖ **Region presets** for East, NE, North, Central, West & South India
- ❖ **Modular pipeline** → add radars, satellites & NWP blending
- ❖ **Open standards:** ODIM, GeoJSON, REST & WebSocket APIs

Challenges & Risks	How We Overcome Them
 No INSAT / lightning feed yet	Adapters built; MOSDAC & IMD/IITM data access
 Radar gaps & outages	Satellite + lightning proxy with confidence score
 Skill drops after 2–3 h	Widening probabilities; NWP blending on roadmap
 AI overload & API keys	Gemini fallback chain → demo mode; keys server-side
 False alarms & trust	Live verification; IMD warnings take precedence

KEY METRICS

* 150-min simulated Nor'wester benchmark, 24 verification samples

0.55

forecast skill (CSI)
at 30 min · 7× baseline*

86%

storm areas caught
at 30 min (POD)*

15 min

new storms flagged
before radar*

<15 s

full 0–6 h forecast
on a laptop CPU

IMPACT AND BENEFITS



Social

Alerts in English & हिन्दी for all



Safety

Early shelter from lightning & squalls



Environmental

Climate-resilient cloudburst alerts



Economical

Protects crops, flights & power



Technological

AI data fusion + live GIS nowcasts

WHO BENEFITS

District & State Disaster Authorities

- ❖ **Targeted Warnings** – **ETA per city & block**, not whole districts
- ❖ **Faster Response** – teams deployed before the storm arrives

Airports & Aviation

- ❖ **Gust & Wind-Shear Alerts** – live countdown for each airport
- ❖ **Safer Operations** – plan holds & diversions in advance

Farmers & Rural Communities

- ❖ **Hail & Cloudburst Warnings** – protect crops & livestock
- ❖ **Lightning Safety** – shelter advice in the local language

NATIONAL IMPACT

Lives & Safety

- ❖ **Fewer Casualties** – lightning & storms kill thousands yearly
- ❖ **Disaster Resilience** – strengthens IMD & NDMA early warnings

Economic Growth

- ❖ **Crop Protection** – timely hail & heavy-rain alerts cut losses
- ❖ **Aviation & Power** – fewer diversions and line outages

Digital Leadership

- ❖ **Mission Mausam** – supports India's weather-ready goals
- ❖ **Open Integration** – GeoJSON, ODIM & REST plug into existing systems

RESEARCH BASE

Nowcasting & Storm Tracking

- ❖ **Radar Extrapolation** – Germann & Zawadzki (2002), Mon. Wea. Rev.
- ❖ **pySTEPS Nowcasting** – Pulkkinen et al. (2019), Geosci. Model Dev.
- ❖ **Cell Tracking** – Dixon & Wiener (1993) TITAN, J. Atmos. Ocean. Tech.

Convection & Hazards

- ❖ **Convective Initiation** – Mecikalski & Bedka (2006) SATCAST, MWR
- ❖ **Lightning Jump** – Schultz et al. (2009), J. Appl. Meteor. Climatol.

Hazards & Data Standards

- ❖ **VIL Density Hail** – Amburn & Wolf (1997), Weather & Forecasting
- ❖ **Downburst Gust** – Stewart (1991) VIL + echo-top relation
- ❖ **Data Standards** – EUMETNET ODIM_H5 radar; ISRO MOSDAC INSAT L1B
- ❖ **IMD Criteria** – rainfall classes, squall & cloudburst (≥ 100 mm/h)

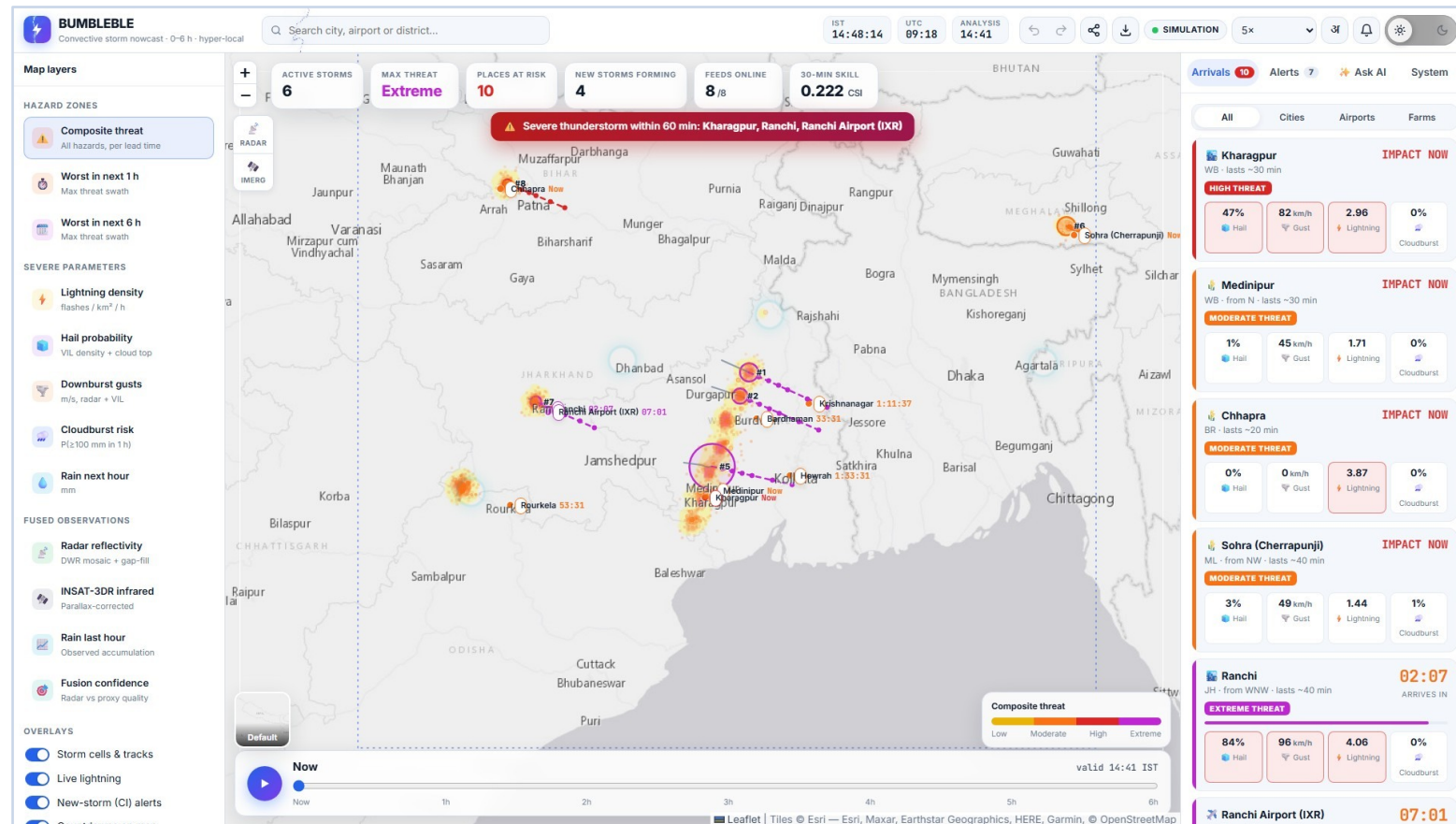
Proven Implementations

- ❖ **IMD DWR Network** – operational district-level nowcasts
- ❖ **NOAA MRMS & UK Met Office** – operational radar nowcasting systems

BENCHMARKING: OUR TECH VS REAL-WORLD SYSTEMS

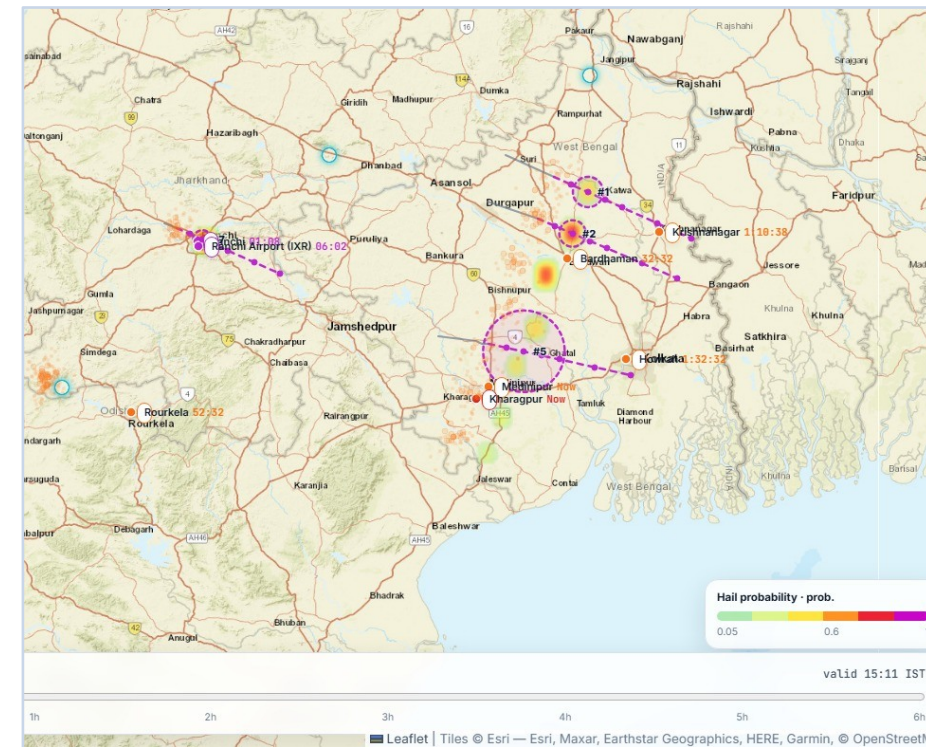
Feature	Our Approach	Real-World Benchmark
Data Fusion	Radar + INSAT + lightning on a 2 km grid	NOAA MRMS multi-radar multi-sensor
Nowcast Engine	Semi-Lagrangian extrapolation + growth/decay	pySTEPS, UK Met Office STEPS
Storm Tracking	40 dBZ cell tracking + lightning jump	NCAR TITAN, NOAA lightning-jump
Initiation Detection	IR cloud-top cooling-rate CI	SATCAST (GOES), EUMETSAT NWC SAF
Warning Delivery	City ETAs, GIS map, AI assistant, exports	IMD district nowcasts, Damini app

LIVE DASHBOARD · SIMULATED NOR'WESTER



- ❖ **Storm tracking** – 6 storm cells with 0–2 h forecast tracks
- ❖ **Live countdowns** – e.g. Ranchi 02:07, Ranchi Airport 07:01
- ❖ **Severe banner** – Extreme-threat cards with hail, gust & lightning
- ❖ **0–6 h timeline** – scrub or animate every 10-min forecast

FORECAST · AI ASSISTANT · PROOF



Hail probability · +30 min forecast

AI assistant · live answer

9/9

automated tests pass

37

forecast steps, 0–6 h

80+

places with countdowns

- ❖ **Verified skill** – CSI 0.55 at 30 min, 7× persistence*
- ❖ **Early warning** – new storms flagged ~15 min before radar*
- ❖ **Real data** – live RainViewer radar frames every 10 min
- ❖ **Deploy anywhere** – one Docker container · free cloud hosting

* 150-min simulated Nor'wester benchmark